M3 · Intro2SE 15031001 · Semester 1/2569 · Group 16

BaanRight

Thai rental posts, turned into decisions — rent, deposit, phone, location and distance from MFU, in one click.

Team **16RideRight** · Chrome extension + web prototype
Repository: github.com/Kennethkaungsatt/BaanRight-mfu-housing-demo- · snapshot `m3-demo`

THE TEAM

16RideRight — who did what

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11 commits · +5,965 / −345 · 33 files

Web prototype, popup redesign, photo tray, docs, slides

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27 commits · +3,903 / −3,427 · 14 files

Extractor logic, test suite, source link, repo cleanup, PR #2

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5 commits · +16 / −5 · 4 files

In-page button, popup labels, photo storage errors, FR-5 bug fix

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0 commits

Non-code work: user interviews (26–27 Aug), testing, translation review

Kanokporn Pakchaipoom
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0 commits

Non-code work: user interviews, Thai copy review, slide content

Numbers from `git log --author=… --numstat`. `kennethape` and `Kennethkaungsatt` are the same account. An account being listed is not a contribution — two members have no commits in the history, and we say so rather than claim otherwise.

THE PROBLEM

MFU students who don't read Thai can't read the rental market

- Housing is advertised in Thai Facebook groups: rent, deposit, phone and soi all buried in free text.
- Translating by hand costs minutes per post — the good rooms are gone before you finish.
- The details that decide a visit are exactly what machine translation mangles: deposit months, soi names, distance to campus.
- Students arrive in year 1 with a dorm, then must find off-campus housing themselves.

A typical post

ห้องให้เช่า ซอยแม่จันใต้ 3
ค่าเช่า 6000 บาท/เดือน
ประกัน 1 เดือน
ห่างจากมหาวิทยาลัย 10 นาที
สนใจโทร 081-234-5678

What the student actually needs

฿6,000 / month · 1 month deposit · 10 min from MFU · 081-234-5678

WHO NEEDS THIS PRODUCT

Primary user, and who we deliberately don't serve

Primary: MFU international students

- Can't read Thai, must rent off-campus after year 1.
- Hunting in Thai-language Facebook groups, often on a phone.
- Needs to answer the landlord in Thai but can't write it.

Secondary

- Exchange / short-stay students arriving mid-semester.
- Thai friends helping a foreign classmate shortlist rooms.

Three moments, one product

1 · Find — paste a post, read rent, deposit, phone, soi and distance in English.
2 · Compare — save listings, search, shortlist with ★, export CSV.
3 · Reply — tick the questions that matter, get a Thai message to paste into Messenger.

Not our user (by design)

Landlords and agents, full-time Thai renters, and anyone wanting a booking or payment platform — those are different products with different requirements.

FUNCTIONAL REQUIREMENTS

5 Must FRs · 16 acceptance criteria — all implemented

ID	REQUIREMENT	ACS	STATUS	WHERE IT LIVES	EVIDENCE
FR-1	Input a housing listing (paste Thai post text)	1	**Working**	Translate tab: textarea, load demo, source link	Paste / demo / source-link pull; empty input rejected
FR-2	Translate a Thai listing into English	2	**Working**	MyMemory client, ≤450-byte chunking, Thai kept on screen	1.5–4 s per demo post; 4 unit tests
FR-3	Extract and save key details: price, deposit, phone, location, availability	8	**Working**	Price hero + details grid; Saved tab; missing field shows "—"	7 extractor tests; save → reload → compare
FR-4	Generate a Thai reply to a landlord	3	**Working**	Ask the landlord → 5 fixed questions → copy	Thai output generated and copied; app never sends (AC-3)
FR-5	Translate selected text directly on a webpage	2	**Working**	`content.js` + context menu, popup opens pre-filled	Real Facebook housing post

Source: `docs/M2-srs.pdf §3` · traceability: `docs/final-report.md §5` (Golden Thread)

NON-FUNCTIONAL REQUIREMENTS

5 NFRs — 4 measured, 1 honestly unverified

ID	KIND	TARGET	HOW WE CHECKED	RESULT
NFR-1	Performance	Translation shown within 10 s	Timed live requests, SRS trials	**Pass** — ≈1.5 s short post, 3–4 s long
NFR-2	Performance	Fields extracted within 5 s	Local extraction is synchronous	**Pass** — sub-millisecond per post
NFR-3	Usability	First use within 30 s, no help	Timed test with an unfamiliar student	**Unverified** — not formally timed
NFR-4	Reliability	Error message, never a misleading translation	Disconnect network, submit; quota + 429 tests	**Pass** — explicit error shown
NFR-5	Security / privacy	Saved listings only in the saver's browser	Profile A saves → profile B opens	**Pass** — no server, no account, B sees nothing

NFR-3 is stated as **unverified** in `docs/final-report.md §7` rather than claimed as passed. Understanding a target and measuring it are different things.

WHAT'S BUILT

Features, on both surfaces

Translate & extract
Chunked th→en translation plus price, deposit, phone, location and availability pulled from free text.

Distance to MFU
Stated minutes, a detected map link, or the Fah Thai market landmark — never an empty cell.

Saved list + shortlist
Search, ★ filter, one-tap call, original post link, photo thumbnails, expandable details.

Photo tray
Add, paste or drag photos; downscaled to ≤1000 px, saved with the listing, downloadable later.

Ask the landlord
Five common questions turned into Thai and copied in one tap — the app never sends it.

Export, alerts, theme
CSV export for comparing, area + budget alerts, light/dark theme, autosaved drafts, PWA offline shell.

The last three are *supporting* features — we don't count them as extra Must FRs, so the scope stays honest. Same workflow in the Chrome extension (inside Facebook) and the web prototype (any device).

SCOPE DISCIPLINE

What we chose *not* to build

- **Auto-searching / scraping Facebook groups** — platform terms; the user pastes or right-clicks instead.
- **Booking, reserving, rental payments** — needs a payment provider and landlord agreements.
- **Contacting landlords automatically** — FR-4 AC-3: the student reviews and sends.
- **Scam / red-flag detection** — an unverifiable claim we can't honour.
- **Legal or contract advice** — outside our competence and a liability trap.
- **Shared database, accounts, login** — needs a backend; privacy by design (NFR-5).
- **Screenshot OCR in this MVP** — excluded by the SRS; the UI says "not in this build".
- **Guaranteeing price or availability accuracy** — the landlord is the source of truth.

15 explicit exclusions in `docs/M2-srs.pdf` §6. Saying no in writing is why the five FRs actually shipped.

TECHNOLOGY

Vanilla JS, two surfaces, no backend

JAVASCRIPT ES2020	≈2,650 lines
HTML5	≈640 lines
CSS3	≈1,410 lines
JSON (MANIFESTS)	2 files
NODE.JS TESTS	112 lines · 11 cases
TOTAL (CODE)	≈5,170 lines

Popup UI
popup.html + css + js

Service worker
context menu, tabs, downloads

Content script
reads the page you're on

Shared core
extractors · translation client · storage

Web prototype
static PWA, offline shell

Not used: TypeScript, React/Vue, bundlers, Node server, database, ORM, auth provider, analytics. Chrome MV3 + a static PWA is enough for this problem — every dependency we skip is one we don't have to maintain.

APIS

One external API. One unused permission. Zero keys.

#	SERVICE	PURPOSE	AUTH / COST	STATUS
1	**MyMemory Translation API**	Thai → English machine translation, post split into ≤450-byte chunks	Free, no key, anonymous fair-use quota	**Used at runtime**
2	Nominatim (OpenStreetMap)	Reverse geocoding	Free	**Declared in manifest, never called**

Chrome APIs: 6 families across 27 call sites — `storage.local` (8), `runtime`, `contextMenus`, `tabs`, `downloads`, `action.openPopup`. Permissions: storage, contextMenus, activeTab, downloads, unlimitedStorage.

Not APIs: Google Maps links are detected by regex inside the post; Facebook is read from the DOM the user is already viewing — no Graph API, no token, no group scraping. The unused Nominatim permission is a known gap we list in the report instead of quietly leaving it.

AI USAGE

No generative model in the product

At runtime

- Translation only, via MyMemory — a machine-translation service that receives text and returns text.
- Extraction is **rule-based** (regex + Thai unit words), as the SRS requires. No LLM decides a price.
- That makes answers deterministic and testable — which is why we can run 11 tests on it.

During development

- **Claude** (27 Aug) — extension development: translation, extraction, storage, feature suggestions.
- **ChatGPT** (27 Aug) — organising the SRS into FRs and NFRs.
- **Codebuff** (9–12 Nov) — M3 deliverables: web prototype, report, diagrams, deck, README, link and asset verification.

Everything AI produced was re-read, syntax-checked and tested before it landed; where output didn't match the requirements we rewrote it (logged in M2 §8). Disclosure: `docs/ai-use-statement.md`. Known weakness: the AI log is per-tool prose, not per-prompt.

GOLDEN THREAD · LIVE DEMO

Four minutes, requirement by requirement

1. **Load demo → Translate** — English plus rent, deposit, phone, location, distance. FR-1, FR-2, FR-3
2. **Point at the details grid** — missing fields show a dash; we never invent a value. FR-3 AC-8
3. **Save → reload → search → ★ → CSV** — compare options side by side. FR-3 AC-7
4. **Ask the landlord → Thai message → copy** — the app never sends it for you. FR-4
5. **Extension:** select Thai text on a Facebook post → right-click → Translate. FR-5

The closing line

"OCR and scam detection are explicitly out of scope in our SRS — the UI says so rather than pretending."

If the network fails on stage

The prototype ships with demo posts and an offline shell; the error path is itself a test (NFR-4), not a crash.

QUALITY

Tested where it matters — 11/11 passing

Automated · `node --test tests/extractors.test`

- Suite loads the **pure functions out of the shipped `popup.js`** — the tested code is the code that ships.
- Deposit formats, Thai digits ๐–๙, floor area vs. location, trailing phone numbers.
- Translation failures: quota text returned with HTTP 200, 429, other statuses.
- **11 tests · 11 pass · 0 fail · ≈75 ms**

Manual

- 2 demo posts × 2 surfaces, 9-step script.
- Real Facebook housing posts in the extension.
- Responsive at 360 / 768 / 1440 px; dark-theme contrast checked.
- Offline, empty input, Ctrl/Cmd+Enter, photo add/paste/dedupe.

Known gaps: FR-5 is manual-only, and there is no CI — tests run locally before a merge.

HONEST LIMITS & NEXT

What isn't there yet

- **Screenshot OCR** — excluded by the SRS; the UI says "not in this build".
- **Translation** — free MyMemory tier, fair-use quota, no offline model.
- **Distance** — the landlord's figure or a landmark estimate, not a routed walk.
- **Storage** — browser only, no account, no sync between devices.
- **NFR-3** — the 30-second first-use target is still unmeasured.

Next

OCR for screenshots · real routing distance from MFU · optional account + sync · Thai UI for landlords.

Before the review

Run the NFR-3 timed test with three first-time users, and remove or use the unused Nominatim permission.

Thank you — questions?

Team 16RideRight · prototype `prototype/index.html` · extension `extension/` · report `docs/final-report.md`